

RateController User Manual

RateController (RC) is a Windows application for controlling the application rate of agricultural products (liquids, granules, seeds, ammonia) in conjunction with AgOpenGPS. It communicates with rate control modules via UDP/Ethernet to manage product application rates, boom sections, and tank quantities.

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1. Getting Started

Before operating, complete these setup steps in order:

1. **Create or load a Profile** — a profile stores all machine settings (products, sections, relays, etc.)
2. **Connect the module** via Ethernet — go to **Modules > Network**, set the subnet, and send it to the module
3. **Configure the module** — go to **Modules > Config** and set the board type and relay type
4. **Configure pins** — go to **Modules > Pins** to verify or set sensor and PWM pin assignments
5. **Configure relay pins** — go to **Modules > Relay Pins** to assign physical pins to each relay
6. **Set valve mode** — go to **Modules > Valves** to select 2-wire or 3-wire valve control
7. **Define sections** — go to **Machine > Sections** and set section widths
8. **Configure relays** — go to **Machine > Relays** and assign relay types to sections
9. **Configure switches** — go to **Machine > Switches** to set master switch behavior
10. **Set up your product** — go to **Products > Rate** and enter product name, control type, units, and rates
11. **Assign the sensor** — go to **Products > Settings** to assign the module ID and sensor ID
12. **Calibrate** — go to **Machine > Calibrate** to determine the Cal Factor
13. **Create a Job** — go to **File > Jobs** to create a job for the field
14. **Connect AgOpenGPS** — the AOG status indicator should turn green

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2. Profiles

Menu: File > Profiles

A profile stores all machine configuration settings. Multiple profiles can be maintained for different implements.

Button	Action
1. New	Create a new profile
2. Load	Load the selected profile
3. Delete	Delete the selected profile
4. Copy	Copy the selected profile to a new name
5. File name	Enter name for New or Copy
6. Files	List of available profiles to select
7. Page Up/Down	Scroll through the profile list

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3. Jobs

Menu: File > Jobs

Jobs track application activity per field. Each job stores application data, notes, and a rate map.

Current Job

Field	Description
Name	Job name
Date	Date/time of the job
Field (1)	Select from list or type a new field name
Notes	Free-text notes for the job

Button	Action
2	Insert current date and time
3	Page up/down through notes

Jobs List

Button	Action
4	Sort jobs by Year
5	Sort jobs by Field
6	Sort filter
7	Delete selected Field
8	Open the selected job
9	Create a new job
10	Copy selected job and its rate map to a new file
11	Delete selected job
12	Page up/down through job list
Show at Start-Up	Display the Jobs menu when the app launches
Activity	Job report
Import	Import jobs
Export	Export jobs

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4. Options – Speed Source & Display

Menu: File > Options

Selects the source of ground speed used for rate calculations and configures display units.

Speed Source

Option	Description
GPS	Use speed from AgOpenGPS
Simulated Speed	Simulate a constant speed (useful for testing)
Wheel Sensor	Use a speed sensor connected to a module GPIO pin

Wheel Sensor Setup

1. Enter the **Module ID** and **GPIO pin** used for speed sensing.
2. Enter the calibration number in **pulses per mile** or **pulses per kilometer**.
3. Press the **up arrow** to send configuration to the module (also sent on Save).
4. Use the **Calibration screen** to calculate the calibration number if unknown.

Display Options

Option	Description
Metric	Toggle between metric (km, ha, L) and imperial (mi, ac, gal) units
Rate Display	Enable the rate display mode on the main screen

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5. Options – CAN

Menu: **File > Options** (CAN tab)

Configures communication with rate control modules via CAN bus. This is an alternative to direct Ethernet/UDP communication.

Enable CAN

Check the **CAN Enable** checkbox to activate CAN communication.

CAN Driver Selection

Driver	Description
SLCAN	Serial port-based CAN adapter. Requires COM port selection.
InnoMaker USB2CAN	Native USB CAN adapter (InnoMaker hardware)
PCAN	Native USB CAN adapter (Peak PCAN hardware)

COM Port (SLCAN only)

When SLCAN is selected:

1. Use the dropdown to select the COM port the adapter is connected to.
2. Press **Refresh** to rescan available COM ports.

Diagnostics

Option	Description
Enable Diagnostics	Enables CAN traffic debug logging

Status Indicators

Indicator	Meaning
Connected	Green — module data received within the last 2 seconds
Driver Found	Green — CAN adapter is open and receiving frames

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6. Products – Rate

Menu: Products > Rate

Configures the fundamental properties of a product.

Setting	Description
1. Product Name	Name of the product
2. Control Type	How the application rate is controlled (see below)
3. Quantity Units	Units used to measure the product (L, gal, kg, lbs, seeds, etc.)
4. Coverage Units	Acre, Hectare, Minute, or Hour
5. Sensor Counts/Unit	Number of sensor pulses per unit of product applied
6. Density	Product density (used for mass/volume conversion)
7. Base Rate	Target application rate
8. Alt. Rate (%)	Alternate rate as a percentage of the base rate
9. Tank Size	Total tank capacity

Control Types

Type	Description
Standard Valve	Uses a valve to vary flow rate
Combo Close	Uses a valve to vary rate and provide on/off control
Motor	Varies motor speed to control application rate
Combo Timed	Uses adjust/pause timing cycles for control

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7. Products – Settings

Menu: Products > Settings

Configures sensor assignment, alarm thresholds, and display preferences per product.

Setting	Description
1. Enable product	Activate this product
2. Rate Sensor Location	Module ID (0–7) and Sensor ID (0–1) for the flow sensor
3. Minimum UPM	Minimum acceptable flow rate — Fixed Value or adjusted By Speed
4. Off-Rate Alarm	% error in rate that triggers an alarm
5. Default Product	Product displayed at startup
6. Scale Weight	Show the scale weight screen

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8. Products – Mode

Menu: Products > Mode

Sets the operating mode for each product. Navigate between products with the left/right arrows.

Mode	Description
Controlled UPM	PID feedback control — adjusts motor/valve to maintain the target rate as speed changes
Constant UPM	Maintains a fixed flow rate based on total machine width regardless of which sections are on. Used for sprayers where turned-off sections return flow to the tank (after the flow meter). RC corrects the measured flow by subtracting the estimated contribution of the off sections to give an accurate application rate per acre.
Document Applied	Tracks and records material that has already been applied — no active control
Document Target	Tracks and records the target rate for comparison — no active control

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9. Products – Data Reset

Menu: Products > Data

Displays accumulated totals for the selected product and allows resetting individual counters. Navigate between products with the left/right arrows.

Counter	Description
Coverage Area 1	Total area covered (primary counter)
Coverage Area 2	Total area covered (secondary counter)
Quantity Applied 1	Total product applied (primary counter)
Quantity Applied 2	Total product applied (secondary counter)
Hours 1	Operating hours (primary counter)
Hours 2	Operating hours (secondary counter)

Press the reset button next to any counter to zero it. Both counters are independent, allowing a session total and a season total to be tracked simultaneously.

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10. Products – Monitoring

Menu: Products > Monitoring

Displays real-time diagnostic data for the selected product. Navigate between products with the left/right arrows.

Rate Information

Display	Description
Target UPM	The current setpoint (desired flow rate)
Applied UPM	The measured actual flow rate
Rate Error %	Difference between target and actual, with sign
PWM	Current PWM output value (0–255)
Frequency (Hz)	Flow sensor pulse frequency

Speed & Coverage Information

Display	Description
Ground Speed	Current speed in MPH or KPH
Working Width	Implement working width in feet or meters
Work Rate	Area covered per hour (ac/hr or ha/hr)

Section Status

Shows 16 section indicator boxes — green for On, red for Off.

Module Communication

Display	Description
WiFi / Signal	Module signal strength (RSSI) bar
Counts per Revolution	Editable sensor calibration value

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11. Products – Control (PID Tuning)

Menu: Products > Control

Configures the PID controller that adjusts the motor or valve to maintain the target rate.

Primary Settings

Setting	Description
1. Gain	Rate adjustment factor. Higher product volume requires lower gain. The effect is exponential — adjust very carefully.
2. Integral	Uses accumulated error to reach target rate faster
3. Max Power	Maximum PWM power delivered to motor or valve
4. Min Power	Minimum PWM power delivered to motor or valve

Adjustment Process

1. **Initial Setup** — Set minimum power to the minimum needed to move the motor/valve. Set maximum power to 100. Set Integral to 0.
2. **Gain Adjustment** — Slowly increase gain until the system overshoots the target, then reduce slightly to stabilize.
3. **Integral** — If needed, increase integral to reach the target rate more quickly.

Optional / Advanced Settings

Setting	Description
1. Deadband	Error % below which no adjustment is made
2. BrakePoint	Error % where adjustment switches to slow rate
3. Slow Adjustment	% of full adjustment to use at slow rate
4. PID loop time	Adjustment loop interval in milliseconds
5. Slew Rate	Maximum total PWM change per PID loop
6. Max Integral change	Maximum Integral PWM change per PID loop
7. Min Start % (timed combo)	% of target rate for a faster start from 0
8. Adjust time (timed combo)	Milliseconds the valve adjusts
9. Pause time (timed combo)	Milliseconds the valve pauses adjustment
10. Min flow sensor Hz	Minimum valid flow sensor pulse frequency
11. Max flow sensor Hz	Maximum valid flow sensor pulse frequency
12. Flow sensor sample size	Number of pulse times used to calculate median Hz

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12. Machine – Calibrate

Menu: Machine > Calibrate

Use this screen to determine the **Cal Factor** (pulses per unit) for a product. The goal is to achieve a consistent meter roller RPM or liquid flow rate matching typical field operation.

Step-by-Step Calibration

Step 1 — Set Ground Speed

- Enter the ground speed you typically operate at.

Step 2 — Set Meter Roller RPM / Valve Position

- Switch the product **ON** using the power button.
- Enter an **Initial Base Rate** and an **Estimated Cal Factor**.
- Press **Start** — RC adjusts flow to match the target rate.
- If successful, RPM is locked.
- The PWM value used is displayed under the lock (for motor controllers).
- RC stops automatically when the target rate is met.

If RC fails to reach target RPM:

- Press **Stop**.
- Check ground speed.
- Adjust the estimated Cal Factor, PWM min/max, or metering drive range.

Step 3 — Set Actual Cal Factor

- Press **Start** to run the meter roller and collect a sample.
- Press **Stop**, then enter the measured amount collected.
- RC calculates and displays the new Cal Factor.
- Press **Save** to record it.

Step 4 — Optional Adjustments

- To redo meter speed: press **Unlock**, then repeat Step 2.
- To reuse a previously saved setting: press **Lock**, then proceed to Step 3.

Important Notes

- AgOpenGPS is **not** used during calibration.
- The Switchbox is **not** used during calibration.
- If a switchbox is present, ensure the **master switch is OFF** before calibrating.
- Calibration uses **Constant UPM, Mode 2**.
- All sections are activated during calibration.
- After calibration, the original mode and section states are restored.

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13. Machine – Sections

Menu: Machine > Sections

Defines the boom sections and their widths.

- There are **16 sections per module** and up to **8 modules**.
- Module 0 = sections 1–16
- Module 1 = sections 17–32, etc.
- Sections can be defined **individually** or grouped into **zones**.
- All sections in a zone share the same width.
- All sections in a zone are controlled by one switch.
- AgOpenGPS can still auto-control each section individually within a zone.
- When creating a new profile, sections must be defined.
- Sections are also automatically redefined when AgOpenGPS section count changes.

Control	Action
Copy Button	Copies the width from Section 1 to all other sections

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14. Machine – Relays

Menu: Machine > Relays

Assigns relay functions to each relay output on the module. There are **16 relays per module** and up to **8 modules**.

Control	Action
Reset (circle)	Reset all relays to default
Renumber (+)	Renumber sections sequentially
Relay Number	The relay output number
Relay Type	The behavior of the relay (see below)
Number	For Switch type: switch number. For Section/Invert_Section type: section number.

Relay Types

Type	Description
Section	Controlled by section switch (AOG, switchbox, or on-screen). Master and Auto switches also apply.
Slave	On when any section relay is on; off when all section relays are off
Master	On when any section relay is on; turns off before section relays turn off
Power	Always on
Invert_Section	On when its assigned section is off
Hyd Up	Hydraulic raise
Hyd Down	Hydraulic lower
Tram Right	Tram marker right
Tram Left	Tram marker left
Geo Stop	Geofence stop output
Switch	Controlled directly by a switch. Not affected by Master or Auto switches.
Invert_Master	On when master is off or all sections are off

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15. Machine – Switches

Menu: Machine > Switches

Configures how the master switch and auto switches behave.

Setting	Description
1. Master Switch – Control All	Controls both the Master relay and all section relays
2. Master Switch – Master Relay Only	Controls only the Master relay
3. Master Switch – Master Override	Master switch is always on (ignores physical switch state)
4. Work Switch	Uses a switch connected to the implement to control On/Off
5. Auto Switch – Rate	Enables automatic rate control
6. Auto Switch – Sections	Enables automatic section control
7. On-Screen Switches – Enabled	Shows on-screen section switches
8. On-Screen Switches – Dual Auto	Shows separate auto rate and auto section buttons on screen

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16. Machine – Pressure

Menu: Machine > Pressure

Calibrates a pressure sensor using two known reference points.

Calibration Procedure

1. Set the system to a **known low pressure**.
 2. Enter the sensor **reading** and the actual **pressure** value.
 3. Set the system to a **known high pressure**.
 4. Enter the sensor **reading** and the actual **pressure** value.
 5. Enter the **minimum sensor reading** that corresponds to 0 pressure.
 6. Press **Show** to display the live pressure screen.
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17. Machine – Primed Start

Menu: Machine > Primed Start

Allows boom priming or a simulated product application while the tractor is stopped. Activate by holding the switchbox master switch in the **On** position for the configured delay time.

Setting	Description
1. On Time	How long the application runs (seconds)
2. Speed	Simulated ground speed during priming
3. Master Switch Delay	How long to hold the master switch On to trigger primed start (seconds)
4. Resume	After On Time expires, if the tractor is moving, continue with normal operation

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18. Modules – Network

Menu: Modules > Network

Connects the application to a rate control module over Ethernet/UDP.

Connection Procedure

1. **(B) Refresh** the IP list to discover modules on the network.
2. **(A) Select** the module's IP address from the list.
3. **(C) Send** the subnet to the module.
 - The **Mod** indicator on the main screen should turn **green**.
 - The module must be connected before sending any other settings.
 - **(E) Select** a board type that matches your module hardware — this populates the **Defaults**.
 - Save changes with **(G)** or cancel with **(F)**.
 - Make any necessary changes on the remaining module pages (Config, Pins, Relay Pins, Valves), saving with **(G)** on each.
 - When all edits are complete, press **(D) Send** to transmit all settings to the module.
 - **(D)** can also be used after single-page edits.

Board Types

Board	Controller
Nano (RC12-3)	Arduino Nano
Teensy (RC11-2)	Teensy 4.1
ESP32 (RC15)	ESP32

Note: After updating module firmware, resend the subnet (step 3) and resend the module config settings (steps 6–7).

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19. Modules – Config

Menu: Modules > Config

Configures the hardware settings of the connected module.

1. Relay Type

Value	Description
0	No relays
1	GPIOs — use micro-controller pins directly
2	PCA9555 — 8-relay module
3	PCA9555 — 16-relay module
4	MCP23017
5	PCA9685
6	PCF8574

2. Pin Assignments by Board

Parameter	RC5 (Nano)	RC8 (Nano)	RC11 (Teensy)	RC11-2 (Teensy)	RC12 (Nano)	RC12-3 (Nano)	RC15 (ESP32)
Sensor Count	2	2	2	2	1	2	2
Relay Control	4	4	1	1	2	4	5
Flow 1	2	2	28	28	D3	3	17
Flow 2	3	3	29	29	—	2	16
PWM 1	5	5	36	36	D9	5	33
PWM 2	9	9	15	15	—	9	26

3. Nano Pin Reference

Pin Name	Number
D0–D13	0–13
A0–A7	14–21

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20. Modules – Pins

Menu: Modules > Pins

Manually assigns the GPIO pins used for sensors and motor/valve control on the module. Pin values range from 0–50. A value of 255 (displayed as "-") means the pin is not assigned.

Setting	Description
Flow Sensor 1 Pin	Input pin for the first flow sensor
Flow Sensor 2 Pin	Input pin for the second flow sensor
Motor Direction 1 Pin	Output pin for motor direction channel 1
Motor Direction 2 Pin	Output pin for motor direction channel 2
PWM 1 Pin	PWM output pin for product 1 motor/valve
PWM 2 Pin	PWM output pin for product 2 motor/valve
Work Pin	Input pin for the work/implement switch
Pressure Pin	Analog input pin for the pressure sensor
Momentary	Enables momentary switch behavior on the work pin

Press **Rescan** to reset all pins to their board-specific defaults (as defined in Config).

Tip: When a board type is selected in Modules > Config, pressing **Defaults** populates these pin values automatically. Only change them if your hardware differs from the standard layout.

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21. Modules – Relay Pins

Menu: Modules > Relay Pins

Assigns the physical GPIO pins to each of the 16 relay outputs on the module. Pin values range from 0–50. A value of 255 (displayed as "-") means the relay pin is not assigned.

- **Low Relays (1–8):** Labeled "Onboard Relays" or "Remote Relays" depending on module config.
- **High Relays (9–16):** Labeled "Onboard Relays" or "Remote Relays" depending on module config.

Press **Rescan** to clear all relay pin assignments.

Tip: When a board type is selected and **Defaults** is pressed in Modules > Config, relay pins are populated automatically for the chosen hardware.

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22. Modules – Valves

Menu: Modules > Valves

Selects the wiring mode for the valve or motor driver.

Option	Description
2-Wire	Simple on/off control — one wire for power, one for ground
3-Wire	Proportional/directional control — direction pin plus PWM output

Select the mode that matches how your valve or motor driver is wired to the module.

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23. Rate Map

Menu: File > Rate Map

The Rate Map enables variable rate application by defining geographic zones with different target rates. It integrates with AgOpenGPS for GPS-referenced mapping.

Zone Management (Zones Tab)

Create zones by clicking **button 1**, then clicking at least **3 points** on the map to define the boundary.

- With **overlapping zones**, the **last zone added** takes priority.
- When variable rate is enabled but **no zone exists**, the **base rate** is used.

Button	Action
1	Create a new rate zone
2	Edit rate zone properties (name, rate per product, color)
3	Delete corner (while creating) or delete zone (while editing)
4	Cancel edits
5	Save edits

Map Display Controls

Button	Action
6 (Window)	Preview window
7 (Enable VR)	Enable / disable variable rate control
8 (Zones)	Toggle rate zones display
9 (Sat View)	Toggle satellite view
10 (Rate Data)	Toggle applied rate display
11 (KML Files)	Toggle KML file display

Data Recording (Data Tab)

Button	Action
1	Data record count
2	Select product applied rate to display (A, B, C, D)
3	Start/stop recording applied rates
4	Delete applied rate data

Variable Rate Look-Ahead (VR Tab)

Configures how far ahead of a zone boundary the application rate begins adjusting, so the rate matches the

new zone's target as soon as the machine enters it. This compensates for the lag in the applicator system responding to a setpoint change.

Setting	Description
Look-Ahead Time (per product)	Seconds before entering a zone boundary that the rate adjustment begins. One value per product (up to 5).
Auto Tune	When checked, RC monitors look-ahead performance and automatically adjusts the look-ahead time to improve zone entry accuracy.

Set look-ahead time manually by entering a value for each product and pressing **OK**. Enable **Auto Tune** to let RC refine the value automatically based on observed transition performance.

Files Tab

Button	Action
Import Zones	Import rate zones from a shapefile
Export	Export the current rate zones to a shapefile (.shp)
Import KML	Import a KML overlay file
Delete KML	Delete a KML file associated with the current job
Import Yield Data	Import yield data into the current job folder

Tip: Long press (right-click) the rate map to drag it to a new position.

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24. Rate Graph

Menu: **Help > Rate Graph** (opens as a separate window)

Displays a real-time chart of the PID control loop for a product. Used for tuning Gain and Integral settings.

Displayed Data

Element	Description
Target UPM line	The desired flow rate setpoint
Applied UPM line	The measured actual flow rate
Error %	Current deviation from target
PWM value	Current PWM output (0–255)
Frequency (Hz)	Current flow sensor frequency

Chart Controls

Control	Description
Auto	Auto-scale the Y-axis to fit the data
Up	Increase Y-axis range by 1.5×
Down	Decrease Y-axis range by 0.67×

The chart maintains the last 30 data points. Click the graph to select which product to display.

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25. Scale / Tank Display

Menu: Products > Settings > Scale Weight (enable checkbox to show)

Displays a floating window showing tank/scale data for a product. Each product has its own scale display window.

Display Modes

Left-click on the displayed value to cycle through four modes:

Mode	Description
Current Weight	Net weight from the scale sensor (after tare)
Applied Weight	Difference between starting weight and current weight
Area Covered	Area worked since the last reset
Application Rate	Applied weight divided by area covered

Controls

Action	Result
Left-click on value	Resets the starting weight and acres reference point
Right-click on value	Resets the tare (zero calibration)
Right-click and drag	Repositions the window

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26. Language Selection

Menu: File > Language

Changes the display language of the application.

Supported Languages

- English
- Deutsch (German)
- Hungarian
- Nederlands (Dutch)
- Polish
- Russian
- French
- Lithuanian

Note: A restart of the application is required for the language change to take effect.

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27. Color Customization

Menu: File > Color

Customizes the foreground and background colors of the application interface.

Preset Color Schemes

Three built-in color presets are available. Select one to apply it immediately.

Custom Color

Select **Custom Color** to open the color picker:

1. Click in the **color gradient panel** to select a hue and saturation.
2. The horizontal position controls **hue**.
3. The vertical position controls **brightness/saturation**.
4. Separate foreground and background colors can be selected.

Color preferences are saved and applied immediately across all screens.

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28. Help – Logs & Diagnostics

Menu: Help (main menu)

Provides access to application logs, module diagnostics, and system information.

Log Files

Log	Description
Activity Log	Records application events and UDP/CAN communication activity in real time
Error Log	Records application errors and exceptions
Ethernet Log	Records raw UDP packet traffic — useful for diagnosing communication issues

Press **Start/Pause** to freeze or resume live log updates.

Module Diagnostics

Display	Description
Module ID	ID of the module currently being viewed (use left/right arrows to switch)
Firmware Version	Firmware version running on the selected module
Elapsed Time	Seconds since last module data was received

System Information

Display	Description
IP / Subnet	Current Ethernet subnet in use
Application Version	Installed RC software version and date
Current File	Name of the active profile file
GPS Coordinates	Live latitude and longitude from AgOpenGPS

External Links

Links to the project repositories are available from this screen:

- AOG RC GitHub repository
- PCB Setup GitHub
- Rate Control PCBs GitHub

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29. Tips & Tricks

Main Screen Navigation

- **To move the main screen**, press and hold in the middle of it, then drag to a new position.
- Use the **minimize button** on the main form to reduce the display to rate only.

Clickable Display Elements

- **Click on current rate** to cycle through rate display modes.
- **Click on target rate label** to switch between the target rate and the alternate rate (a percentage of the target). When variable rate is enabled, the label shows the VR target rate and is not clickable.
- **Click on quantity applied label** to switch between quantity applied and quantity remaining.
- **Click on area applied label** to switch between area applied and area remaining.
- **Click on area applied value** to reset the area counter.
- **Click on the tank bar graph** to reset the tank quantity.
- Click the **rate graph** to select which product to display.

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30. Troubleshooting

No Relays Activating

1. Is the module Config set correctly for board type and relay type?
2. Has the subnet been set in the app and uploaded to the module?
3. Have section count and widths been set?
4. Is Auto mode on?
5. Is a section assigned to each relay?
6. Is a switch assigned to each section?
7. If Auto is on, have sensor counts/unit and base rate been set?

No UPM (No Flow Reading)

1. Check ground speed.
2. Check working width.
3. Check sensor control settings.
4. Is AgOpenGPS connected?
5. Is the module connected?
6. Check sensor counts/unit.
7. Check base rate.
8. Have control values been sent to the module?
9. Check control type — is it a valve or motor?
10. Is the connector fully plugged in?
11. Is the module running the latest firmware?
12. Is Auto mode on?

No Rate Adjustment

1. In manual mode, the master switch must be on.
2. Is minimum PWM too low?

Motor Won't Shut Off

1. Minimum PWM should be set to 0.
2. Is master override enabled?

After Updating Module Firmware

1. Resend the subnet (Modules > Network).
2. Resend all module config settings (Modules > Config, then Send).

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